

Part I:

1) Descriptive Statistics [HW 1.1, 1.2]

- Location: Mean, Median
 - Variability: Range, Variance/Std. Deviation
- } And how they change under linear transformation.

Book Chapters: 1.1 (only the beginning), 1.3, 1.4 (note - a different def. of variance...)

Things to Know:

- Understanding and computing different descriptive stats.

2) Set Theory [HW 1.3]

- Experiment, Sample Space, Event
- Sets: Empty Set, Subsets, Equality of Sets
- Operations: Union, Intersection, Complement
- DeMorgan's Law
- Properties of two (or more) sets: Disjoint, Mutually Exclusive, Collectively Exhaustive

Book Chapters: 2.1

Things to Know:

- Computing Set Operations
- Proving one set is a subset of some other set.

3) Fundamentals of Probability [HW 1.4, 1.5, 2.1, 2.2, 2.3]

- 3 Axioms of Probability, Properties (e.g. Law of Total Probability)
- Equally Likely Outcomes and Counting Techniques
 - + Product Rule for "Sub-Experiments"
 - + Permutations and Combinations
- Conditional Probability and Bayes' Rule
- Independence of 2 Events, Pairwise / Mutually Indep. Events

Book Chapters: 2.2, 2.3, 2.4, 2.5

Things to Know:

- Using Axioms of Prob. to compute the prob. of different events
- Counting the # of outcomes in a: i) Sample Space, ii) Event
- Computing Conditional Probabilities, and using Bayes' Rule

4) Discrete Random Variables [HW 2.4, 2.5, 3.1, 3.2, 3.5]

- Def. of a R.V., relation to a sample space
- Characterization: Prob. Mass Fn., Cumulative Dist'n Fn.
 - + Interpretation and Properties
- Expected Value of: X , $g(X)$. Linearity of $E[\cdot]$
- Variance (of a r.v.): Def'n and Properties. Standard Deviation
- Common Families of Discrete RV's: PMF, Parameter(s), Interpretation, Properties
 - + Bernoulli, Geometric, Binomial, Poisson

Book Chapters: 3.1, 3.2, 3.3, 3.4, 3.6 (only the first part)

Things to Know:

- Finding and working with pmf's/cdf's
- Computing $E[\cdot]$ and $\text{Var}(\cdot)$ for a discrete r.v. or a function of a discrete r.v.
- How to identify and work with common distributions

5) Continuous Random Variables [HW 3.3, 3.4]

- How they differ from discrete r.v.'s
- Characterization: Prob. Density Fn., Cumulative Density Fn.
 - + Interpretation and Properties
- Expected Value of: X , $g(X)$. Linearity of $E[\cdot]$
- Variance: Def., Properties. Standard Deviation
- Common Families of Cts. R.V.'s: PDF, Parameter(s), Interpretation, Properties

+ Uniform, Exponential, Normal / Gaussian

Book Chapters: 4.1, 4.2, 4.3, 4.4 (first part)

Things to Know:

- Finding and working with pdf's/cdf's
- Computing $E[\cdot]$ and $Var(\cdot)$ for a cts. r.v. or a function of a cts. r.v.
- How to identify and work with common distributions

6) Jointly Distributed Random Variables [HW 4.1-4.5]

• Discrete RV's:

+ Joint / Marginal / Conditional PMF's (only if 2)

- Definitions, Properties

+ Independent RV's (Mutual Indep. if more than two)

• Continuous RV's:

+ Joint / Marginal / Conditional PMF's (only if 2)

+ Independent RV's (Mutual Indep. if more than two)

• Expectation w/ Joint RV's

+ Linearity; $E[X_1, \dots, X_n]$ and $Var(X_1 + \dots + X_n)$ under independence

• Covariance: Def'n, Properties

+ Linearity, $Var(X_1 + X_2)$ without independence

+ Correlation: a normalization of Covariance

• Conditional Expectations: Def'n and "Tower Rule" || Not on midterm

Book Chapters: 5.1, 5.2 (first part)

Things to Know:

- Checking if a joint pmf/pdf is valid
- Computing Marginal / Conditional pmf/pdf's
- Checking independence; or building a joint pmf/pdf from indep. r.v.'s
- Computing Expectations - including covariance and conditional exp.'s

Some Useful Tools:

- Product Rule for Sub-experiments + Counting
- Manipulating expressions like " $P(c \leq aX + b \leq d)$ "
- $E[aX + bY] = a \cdot E[X] + b \cdot E[Y]$ for any X, Y (not necessarily independent)
- $Var(aX + b) = a^2 \cdot Var(X)$
- $Var(aX + bY) = a^2 Var(X) + b^2 Var(Y)$ if X, Y are independent
- $Cov(aX + b, cY + d) = a \cdot c \cdot Cov(X, Y)$